# How to implement UP or Down Counter by Combinations of Clock & Outputs

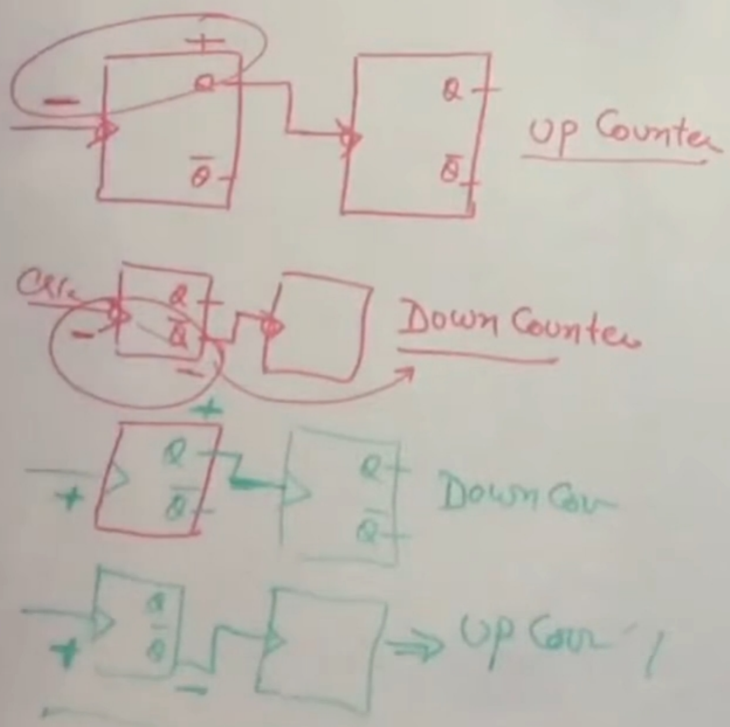
## 3-Bit UP-DOWN RIPPLE COUNTER

$\boxed{- \; Q + ve}$ — Up

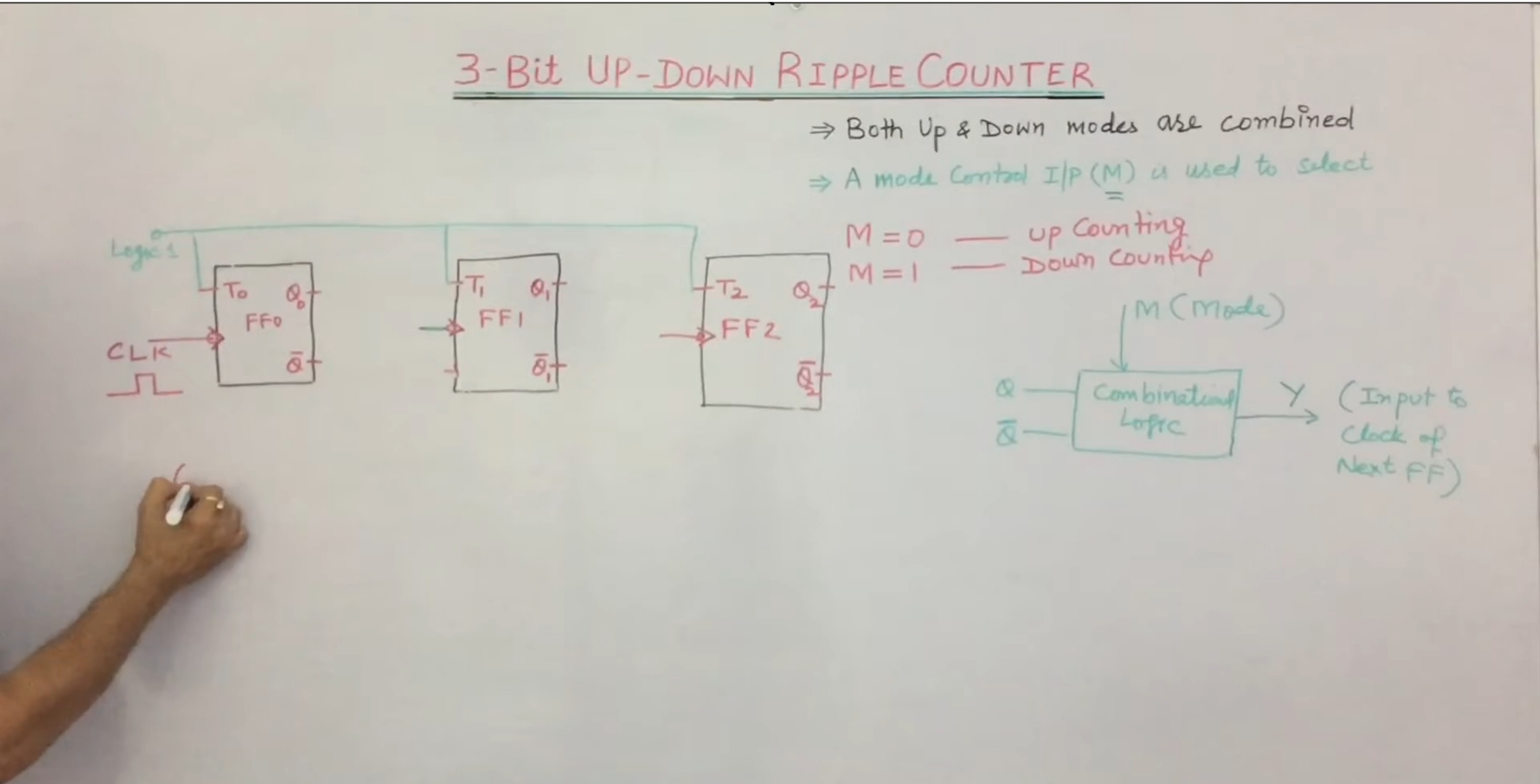
1. −Ve Edge Trig FF → Q is clock = UP Counter
2. −Ve Edge Trig FF → $\bar{Q}$ is clock ⇒ Down Counter
3. +ve Edge Trig FF → Q is clock ⇒ Down Counter
4. +ve Edge Trig. FF → $\bar{Q}$ is clock ⇒ UP Counter

⇒ Both Up & Down modes are combined

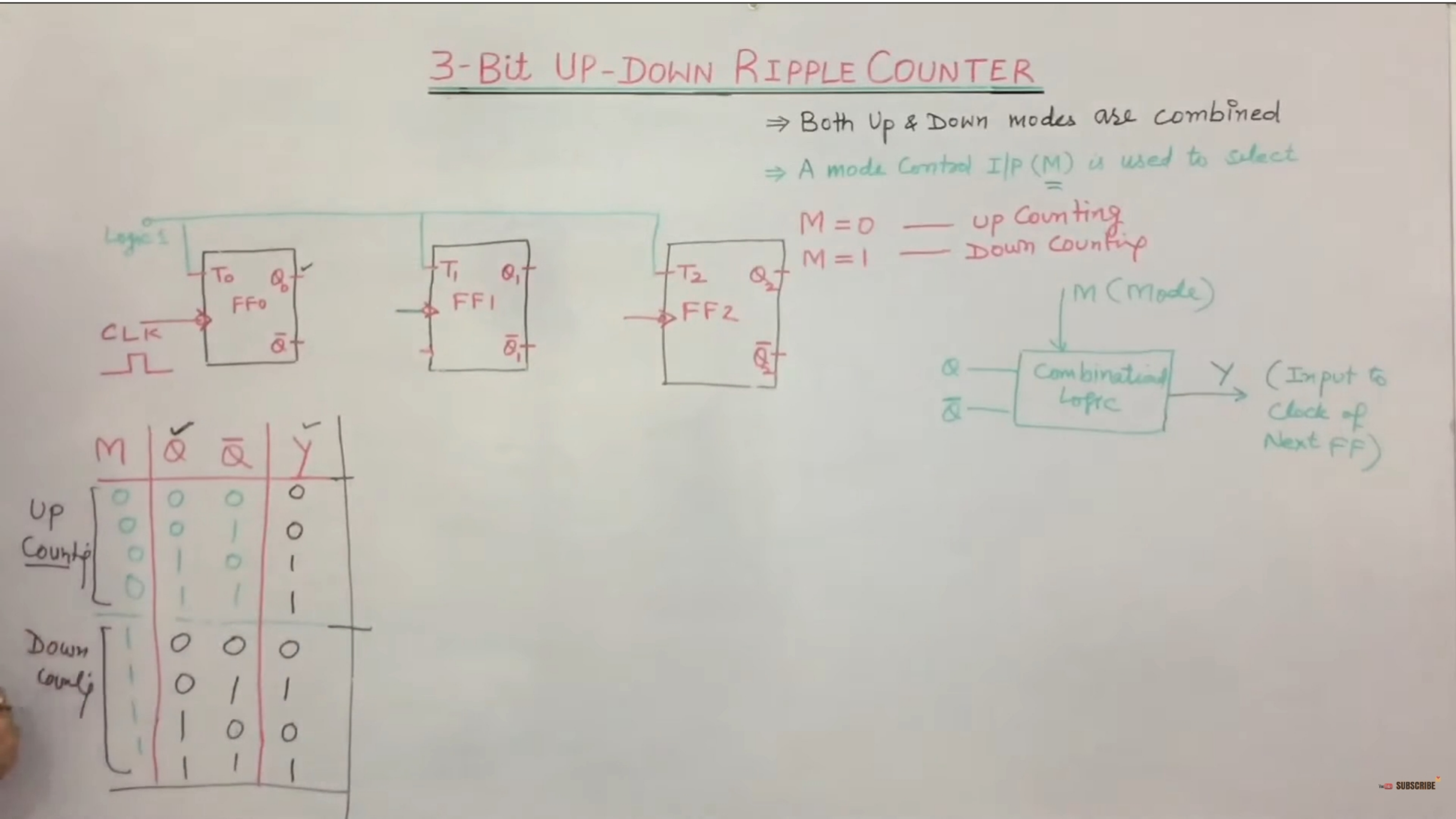
⇒ A mode Control I/P (M) is used to select



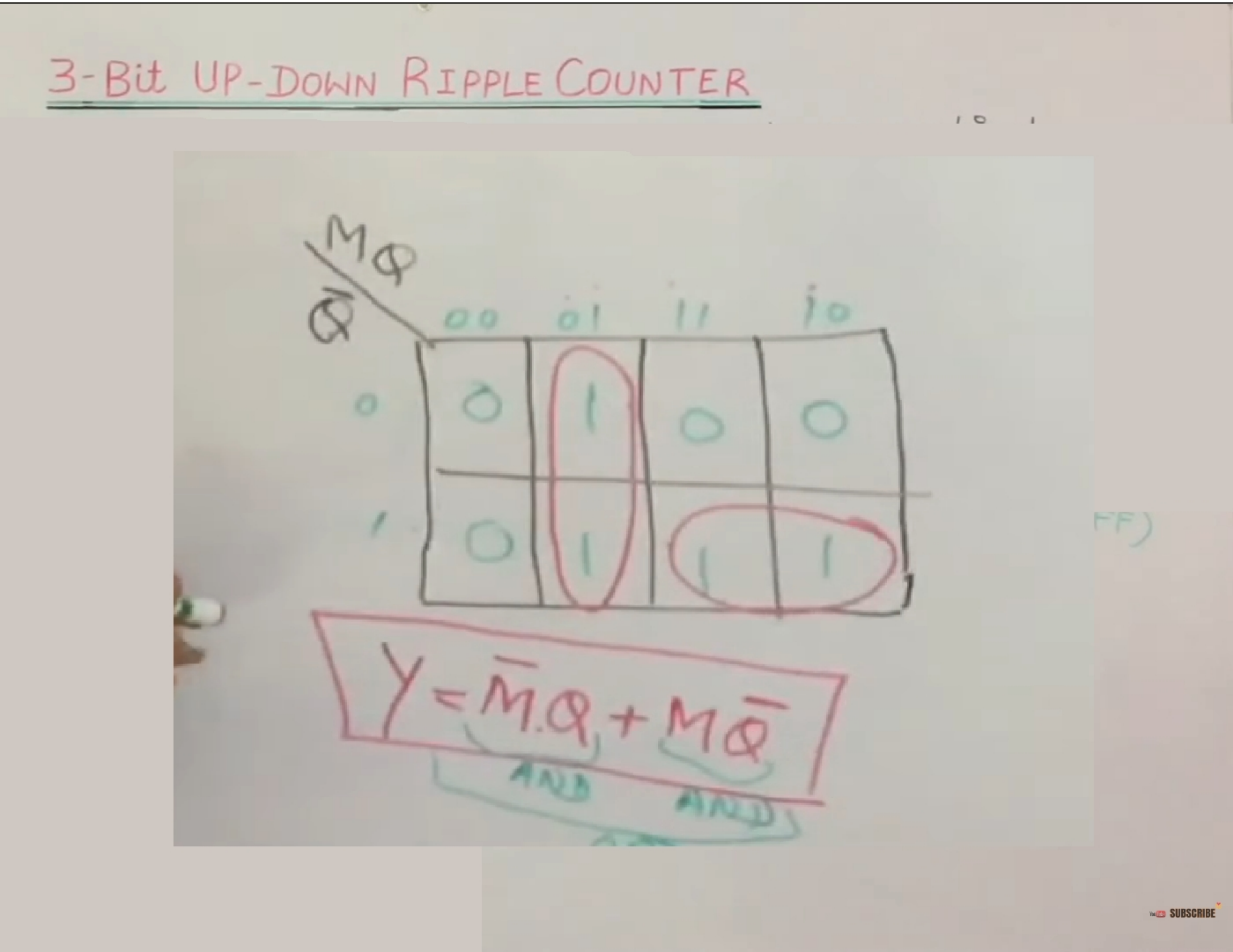
# Block diagram in which we will insert Logic of I/P M :-



# State Diagram for M along with Q & $\overline{Q}$ inputs & Output Y :-



# K-Map for input M along with Q & Q :-



# Final Ckt. for 3-bit Up & Down Async. counter :-